



## Research Paper

## Decision-making autonomy and depressive symptoms among women in Lesotho: Rural–urban differences

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## ARTICLE INFO

## Keywords:

Depression  
Decision-making autonomy  
Pregnancy loss  
Rural–urban disparities  
Lesotho  
Spatial analysis

## ABSTRACT

**Objectives:** To assess the association between depressive symptoms and decision-making autonomy with rural urban inequalities, and mapped province-level spatial disparities in depression prevalence in ever-married women in Lesotho.

**Methods:** This study used cross-section data of 3297 women aged 15–49 years from the 2023–24 Lesotho Demographic and Health Survey (LDHS). Depression was measured using the Patient Health Questionnaire-9 (PHQ-9  $\geq 10$ ). Women's decision-making autonomy was measured using a composite empowerment index based on participation in five household decision-making domains. Survey-weighted multivariable logistic regression models were fitted at the national level and stratified by place of residence. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Spatial analyses were conducted to examine district-level variation in depression prevalence and autonomy.

**Results:** Overall, 7.4% (95% CI: 6.3–8.7) of women reported moderate-severe depressive symptoms. Women with decision-making autonomy showed lower odds of depression nationally (AOR = 0.63; 95% CI: 0.39–0.98), with a stronger effect in rural areas (AOR = 0.44; 95% CI: 0.27–0.74). Depression prevalence was highest in central–northern districts (Maseru, Berea, Leribe, Butha-Buthe) and lower levels of autonomy observed in southern and south-western provinces.

**Conclusions:** Women's decision-making autonomy is significantly associated with lower odds of depressive symptoms in Lesotho, particularly among rural women. The observed rural–urban and spatial inequalities highlight the need for context-specific mental health and reproductive health interventions that incorporate women's empowerment as a core component.

## 1. Introduction

Depression is one of the causes of disability in the whole world and it is even higher among women in reproductive age (Herrman et al., 2022). Depression is more common and prevalent in women than men because of an aggregate of biological, social, and environmental issues (Patel et al., 2018). Social roles are gendered, and access to economic resources is limited in low-resource settings, making people more vulnerable to mental health problems due to reproductive roles and restricted access to mental health (Patel et al., 2018).

Women possessing depression do not only impact the quality of life, but the functioning of the family, child health, and well-being of the community, in general (Oram et al., 2022). It is thus important to understand the social determinants that affect the depressive symptoms in

order to bring meaningful, context-sensitive intervention (Amouzou et al., 2025). The decision-making autonomy of women is one of the important social determinants of mental health (The Lancet, 2025).

Decision-making autonomy is defined as a woman having power to engage in making decisions regarding her health care, family purchases, finances, mobility and reproductive decisions (Heise et al., 2019). The results of numerous nations show that women who have more freedom are less likely to encounter depressive and anxiety symptoms (Westby et al., 2021). Autonomy can keep mental health safe because it would enhance the sense of control that women have in their lives, decrease their vulnerability to coercive relationships, and allow access to health and social resources (Leight et al., 2022). On the other hand, a lack of autonomy has been associated with higher psychological distress, helplessness and increased vulnerability to common mental

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<https://doi.org/10.1016/j.jadr.2026.101080>

Received 22 January 2026; Received in revised form 5 April 2026; Accepted 9 May 2026  
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disorders (Leight et al., 2022; Miah, 2025). Although the increased awareness of the need of autonomy among women and the mental well-being of women has advanced throughout the globe, few studies have been conducted in most low-income countries (LMICs) (Borges do Nascimento et al., 2025; Jia et al., 2025). Mental health of women in sub-Saharan Africa is a product of socioeconomic, reproductive, and contextual risk factors (Bandara et al., 2024). There are early marriage, early childbearing, internet use, pregnancy loss, high fertility and large family size, which are usually stressful factors that can make an individual vulnerable to depression (Miah and Ullah, 2025). These risk factors are usually accompanied by limited decision-making autonomy especially in rural regions where traditions and gender norms are the order of the day (Rahman et al., 2024). Although urban setting usually has more access to education, work, and health services, other stressors, such as economic insecurity, social isolation, and accelerated sociocultural change, might also be brought about by urban setting (Nabaggala et al., 2021). These aspects indicate that the interdependence between autonomy and depression might depend on the setting and there is a necessity to explore the rural-urban variations in mental health outcomes (Acharya et al., 2010b, 2010a; Idris et al., 2023; Mustafiz et al., 2025).

In Southern Africa, in Lesotho, depressive symptoms remain a major public health issue, with recent national surveys reporting a substantial burden of untreated mental health conditions among women, driven by digital access, poverty, human immunodeficiency virus (HIV) prevalence, and limited access to mental-health services (DHS Program, 2023; Jejaw et al., 2026). Additionally, Lesotho showed the highest estimated prevalence of depression, with women excessively suffered (Myint et al., 2025). Likely, a slightly higher prevalence of women sought (17.1%) help for depression compared to men (16.4%). Particularly, women's living in urban's settings showed higher help-seeking prevalences compared to men (Fernández et al., 2025; Okyere et al., 2025). Previous studies also shows that women's autonomy in Lesotho remains remarkably inadequate, mainly in decisions related to health care, household purchases, and mobility, with rural women experiencing even lower levels of intervention due to entrenched gender norms and economic dependence (Gebeyehu et al., 2022; Hormazabal-Salgado et al., 2024; Viens et al., 2016). In Lesotho, limited studies exist about the mental health of women and their social determinants (DHS Program, 2023). There has been a preponderance of studies which have concentrated on maternal and reproductive health outcomes and little has been done on psychological well-being (Leseba et al., 2025). Naturally, there is scanty evidence on the association between decision-making autonomy with depressive symptoms among women in national representative samples (Mokwena and Makhozonke, 2024). In addition, the rural-urban disparities or province-wise spatial inequalities in the prevalence of depression have not been investigated in previous studies, which are important in determining the areas of high burden and effective intervention strategies (Okyere et al., 2025). This gap in evidence is a major shortcoming since there is a need to combine policy and programmatic planning in Lesotho to address the issues of women empowerment and mental health (Myint et al., 2025).

Theoretically, women's autonomy influences mental-health outcomes through multiple mechanisms. Guided by Self-Determination Theory and Gender and Power Theory, limited autonomy reduces women's capacity to make independent decisions regarding health care, household resources, and mobility. Such constraints increase psychological stress, diminish perceived control, weaken coping ability, and ultimately elevate the risk of depressive symptoms (Hindman et al., 2026; Raz et al., 2025). These theoretical perspectives support the link between autonomy and depression in the context of Lesotho (DHS Program, 2023). Against this background, the present study aimed to examine the association between decision making autonomy and moderate-severe depressive symptoms among women in Lesotho using nationally representative LDHS data (DHS Program, 2023). Specifically, this study assessed this association at the national level, conducted

stratified analyses by rural and urban residence to explore inequalities, and mapped province-wise spatial variation in depression prevalence. By integrating individual-level regression analyses with spatial methods, this study seeks to provide policy-relevant evidence to inform targeted mental health and reproductive health interventions in Lesotho.

## 2. Methodology

### 2.1. Setting, study design and source of data

This paper is based on a secondary analysis of the data of the 2023–24 Lesotho Demographic and Health Survey (LDHS), a household survey that was conducted nationally by the Lesotho Ministry of Health with technical assistance of ICF under the DHS Program. The data was collected from 27 November 2023 to 29 February 2024. To present current estimates of the demographic and health variables, such as fertility, maternal and child health, and mental health according to the Sustainable Development Goals (SDGs). The LDHS utilized standardized DHS questionnaires (household, women, men and biomarker) that were translated to the Lesotho setting and were conducted in the Sesotho and English languages. Face to face interviews were conducted using tablet-based computer aided personal interviewing (CAPI) and inbuilt quality checks were used (DHS Program, 2023). This study followed Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines (Cuschieri, 2019).

### 2.2. Sampling design and population of study

LDHS used a stratified cluster sampling design, which was two-stage using the 2016 Lesotho Population and Housing Census. At the first level, 400 enumeration areas (EAs) were chosen on a basis of probability proportional to size. Twenty-five households were sampled per EA using systematic random sampling as per the second stage. All eligible persons were all women aged 15–49 years, and were usual residents of the selected households, or had stayed in the selected households the night before the survey. This was an analysis of women who were on subsampled households and who had completed PHQ-9 depression module. The weighted analytic sample was the final sample of 3, 297 ever-married women aged 15–49 [Fig. 1].

### 2.3. Outcome variable

The main outcome variable Patient Health Questionnaire-9 (PHQ-9) was used to evaluate the depressive symptoms. A PHQ-9 score of 10 and above was considered a moderate-severe depressive symptom in women whereas a score below 10 was considered no-depression. All regression analysis used this binary outcome. Internal consistency of the PHQ-9 in this sample was assessed using Cronbach's alpha ( $\alpha = 0.82$ ), indicating good reliability. (Miah and Ullah, 2025; Westby et al., 2021).

## 3. Exposure variable decision-making autonomy

The main exposure variable was decision-making autonomy which was measured by a composite index of empowerment based on five domains of household and reproductive choices:

1. Own health care.
2. Large household purchases.
3. Use of household money.
4. Visits to relatives or family.
5. Family planning decisions.

Women were regarded autonomous in each domain if they reported making the decision alone or jointly with their partner. Each empowered domain contributed 1 point, resulting in a composite score from 0

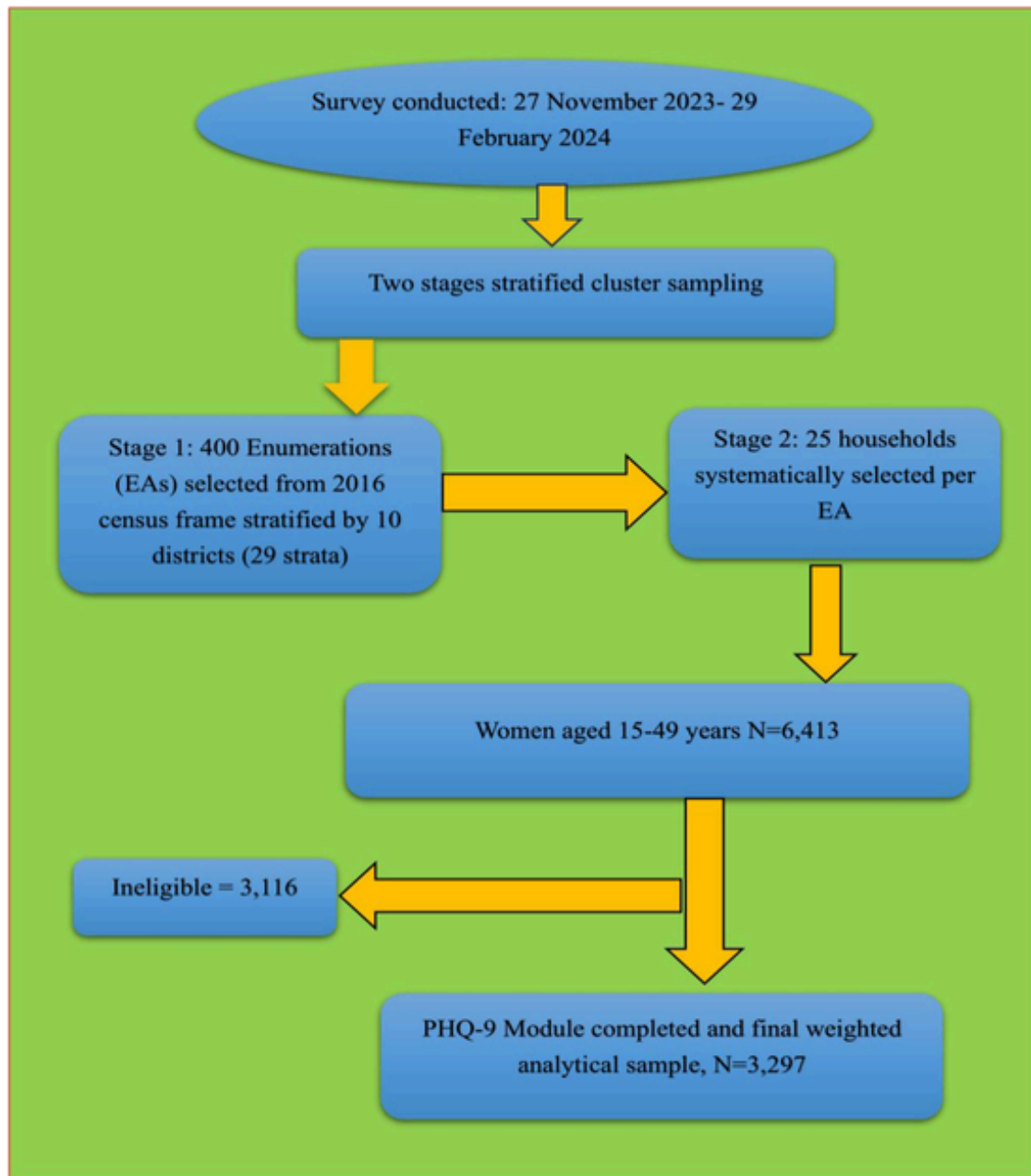


Fig. 1. The Study Flow.

to 5. Then, this study considered, no autonomy: composite score  $\leq 2$  and yes autonomy: composite score  $\geq 3$ . This binary variable (Decision-Making Autonomy) was used as the main exposure in all analyses examining its association with depressive symptoms. This method corresponds with the past research conducted by the DHS on the level of empowerment of women and represents the meaningful involvement in the household decision-making process. This binary measure captures the overall level of meaningful involvement in household and reproductive decisions, providing a clear indicator of women's autonomy for analysis (Ameyaw et al., 2016; Anfaara et al., 2024; Antabe et al., 2025; Kamiya, 2011; Miah, 2025).

### 3.1. Covariates

In this study, women's sociodemographic characteristics were categorized in line with standard DHS methods. Women's age was categorized into seven groups (15–19, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49 years), while age at first birth was grouped as early (10–17 years), normal (18–24 years), and late ( $\geq 25$  years). Age at first sexual intercourse was categorized as  $< 18$  years or  $\geq 18$  years, and age at cohabitation was classified into young ( $< 18$  years), mild (18–20 years), and older ( $\geq 21$  years). Women's education was categorized into below primary, secondary incomplete, and secondary or higher, and maternal employment was categorized as currently working or not working. Adequate antenatal care (ANC) visits were dichotomized as  $< 4$  or  $\geq 4$  visits. Reproductive characteristics included number of children (none, 1–2, 3+), birth size (small, average or large), preterm birth (term  $\geq 9$

months, preterm <9 months), current pregnancy status (yes/no), pregnancy multiplicity (single/multiple), pressure to be pregnant (yes/no), and history of pregnancy loss (none, one loss, two or more losses). Residing status was classified as residing with or not with husband/partner, current abstinence as yes/no, number of unions as once or more than once, and recent menstruation as yes/no. IPV justification (does not justify/justifies IPV). Household environment and socioeconomic factors included place of residence (urban/rural), administrative province (Butha-Buthe, Leribe, Berea, Maseru, Mafeteng, Mohale's Hoek, Quthing, Qacha's Nek, Mokhotlong, Thaba-Tseka), religion (Roman Catholic/Other), wealth status (poor, middle, rich), household size (1–3, 4+ members), household head sex (male/female), household assets (ownership of motorbike or bicycle; yes/no), media access (access to TV, radio, newspaper, or mobile; yes/no), and household construction materials (based on roof materials; improved/unimproved). Source of drinking water and sanitation facility were also categorized as improved or unimproved. Internet use in the last 12 months was coded as yes/no. All covariate categories were harmonized with previous DHS-based studies to ensure consistency in reporting and comparability across surveys and analyses (Esteban-Gonzalo et al., 2026; Miah et al., 2026; Tamanna et al., 2025; Tamanna and Mahmud, 2025; Zoha et al., 2025).

### 3.2. Data quality and management

The data were entered electronically (CSPRO) and checked automatically in terms of range and consistency. Supervision of data on a daily basis, transfer in safe data and central data cleaning guaranteed high data quality. Additional checking and cleaning of secondary data were performed before final analysis.

### 3.3. Statistical analysis

All the analyses factored in the complicated survey design with sampling weights, clustering, and stratification in order to make it representative of the nation. An analysis of complete-cases was applied.

## 4. Code availability

The cleaned dataset, analysis scripts, and related resources supporting the findings of this study are available at GitHub:

<https://github.com/muhammadsalek/Rural-Urban-Inequalities-in-Depression-among-Ever-Married-Women-in-Lesotho>. The analyses were conducted following standard public health and epidemiological approaches and are consistent with methods applied in previous literature (Celentano ScD MHS and Szklo MPH DrPH, 2019).

### 4.1. Descriptive, and bivariate analysis

The characteristics of the participants were summarized using descriptive statistics. Chi-square tests were used to test bivariate relationship between depressive symptoms and explanatory variables. Multivariate modeling was considered on variables having  $p < 0.20$  or good conceptual relevance (Celentano ScD MHS and Szklo MPH DrPH, 2019). Spatial analysis to depict spatial difference in the prevalence of depression and autonomy of women in decision-making at the province level, choropleth maps were shaped.

### 4.2. Multivariable modeling

Logistic regression was used to assess associations between depressive symptoms and decision-making autonomy. Models were built sequentially using a forward stepwise approach, informed by both statistical significance and prior literature:

- **Model 0:** Maternal age and decision-making autonomy variable just.
- **Model 1:** Sociodemographic factors (education, religion, residence).
- **Model 2:** Reproductive factors (age at first birth, cohabitation, parity, multiple pregnancy, pregnancy loss).
- **Model 3:** Contextual factors (urban/rural residence)
- **Model 4 (Final model):** Household and environmental factors (toilet facility, water source, household size, household head sex, wealth, media and internet access, household assets and materials).

The final model was nominated based on conceptual relevance and improved model fit (AIC = 1648.22; BIC = 1886.15; mean VIF = 2.2).

## 5. Final model adjusted covariates

This set of confounders were adjusted in the final balanced model including women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials.

### 5.1. Model diagnostics

Multicollinearity was assessed using variance inflation factors (VIF). Model fit was assessed using the Hosmer–Lemeshow test, which showed good fit ( $\chi^2 = 5.85$ ,  $p = 0.6642$ ). Model discrimination was assessed using the area under the ROC curve (AUC = 0.6781), demonstrating moderate predictive ability. Results are reported as adjusted odds ratios (aORs) with 95% confidence intervals, with statistical significance set at  $p < 0.05$ .

## 6. Sensitivity analysis

In this study, no additional sensitivity analyses were conducted for the PHQ-9 scores, as we adhered to the standard DHS scoring protocol for defining moderate-severe depressive symptoms (DHS Program, 2023).

### 6.1. Ethical considerations

The LDHS protocol acknowledged ethical approval from relevant ethics committees in Lesotho and the ICF Institutional Review Board. Written informed consent was obtained from all participants. This study used anonymized, publicly available DHS data with permission from the DHS Program. However, details available at <https://dhsprogram.com/Methodology/Protecting-the-Privacy-of-DHS-Survey-Respondents.cfm>.

## 7. Results

The prevalence of depression was 7.4% (95% CI: 6.3–8.7). On place of residence, just over half (54.98%) respondents reported the rural (95% CI: 51.47–58.45) and 45.02% reported the urban area (95% CI: 41.55–48.53). The small overlap between the confidence limits implies that more rural participants are represented in the study population though, to establish the statistical significance of the difference, this would have to be tested. The confidence intervals of the two categories of depression and non-depression are complementary as expected, and it proves that depression is a minor yet non-trivial burden to the population health [Figs. 2, 3].

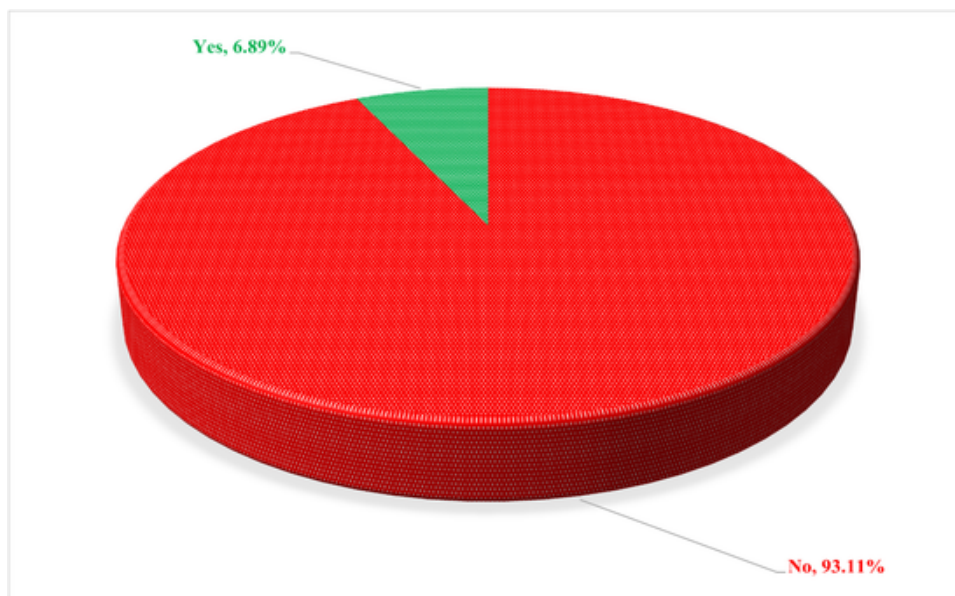


Fig. 2. Prevalences of Depression Symptoms.

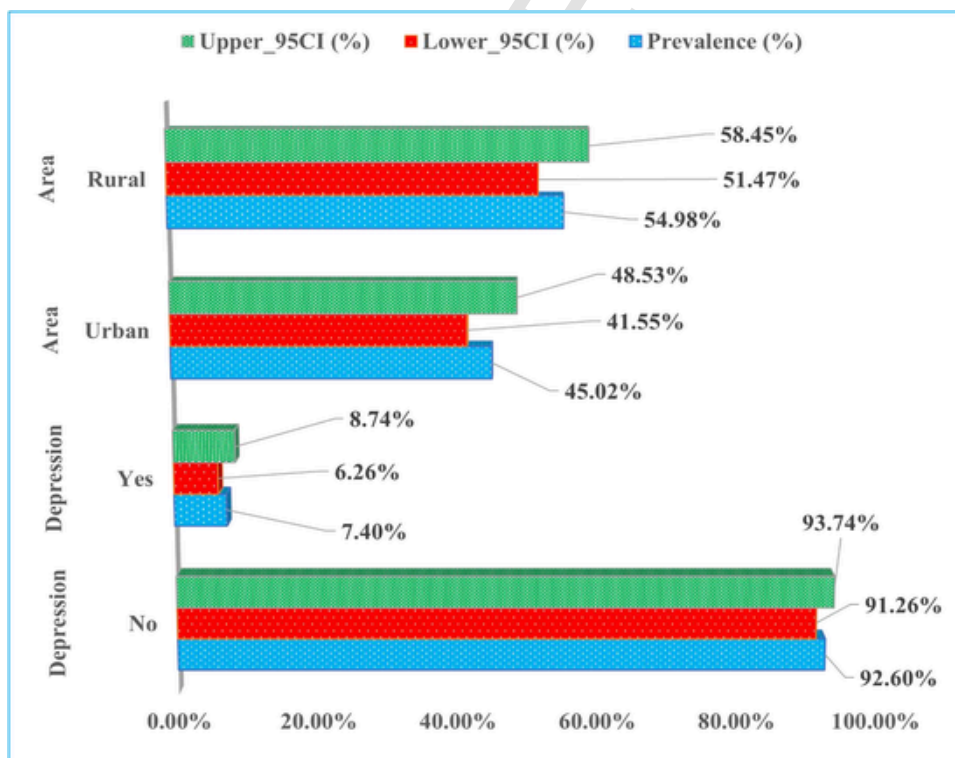


Fig. 3. Rural-Urban Inequalities of Depression Prevalence.

The prevalence of the rural population in the study population was 54.98% (95% CI: 51.47–58.45) and that of the urban population was 45.02% (95% CI: 41.55–48.53). The fact that the CIs are mainly non-overlapping implies that the percentage of rural members might be higher than that of urban people, yet it is necessary to test whether the difference is significant. The depression prevalence was 7.40% (95% CI: 6.26–8.74) meaning it is possible to suggest that depression can be ex-

perienced by 6–9% of the target population. On the other hand, 92.60% respondents were not depressed (95%CI: 91.26–93.74), which is an accurate estimate since the percentage in this group was large. All in all, these results suggest that the prevalence of depression is lower than in the majority of the population, and rural inhabitants are a slightly higher proportion of the study sample [Fig. 4].

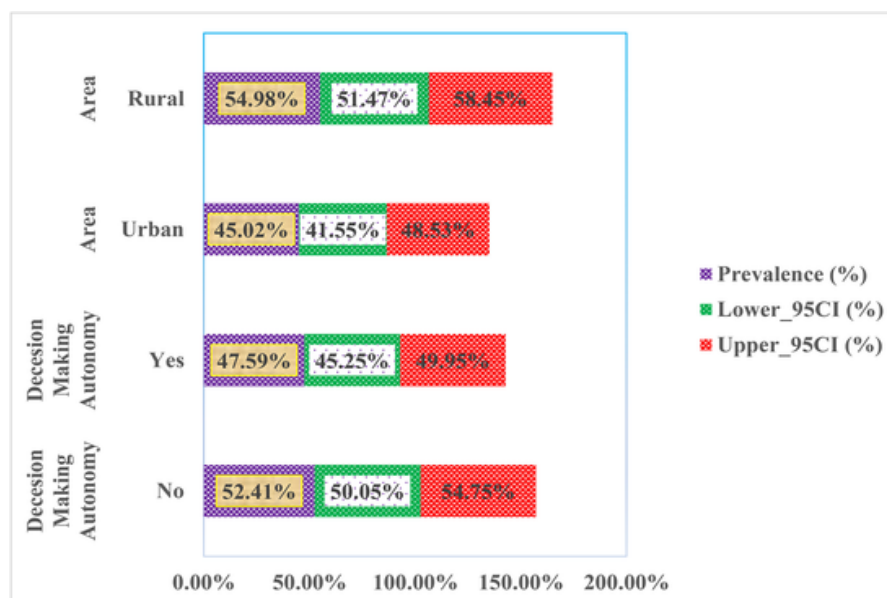


Fig. 4. Rural-Urban Inequalities of Decision-Making Autonomy.

On the whole, depressive symptoms were observed only in a few prevalences of women. Particularly, higher among women in the age groups of 15–19 years, 25–29 years, and 35–39 years. On the other hand, early reproductive such as early fertility, early onset of sexual activity, and early cohabitation were associated with relatively high levels of depressive symptoms. The variations due to socioeconomic status were relatively low, with women having lower education levels and being unemployed displaying relatively high prevalence. In relation to health determinants, limited variation was noted apart from women who had relatively fewer antenatal visits and women with high parity having a higher prevalence of depressive symptoms. Women who lacked decision-making power and those from urban centers had relatively high depression levels. The variations by district were also significant [Table 1].

When controlling the odds of depression against a comprehensive set of sociodemographic, reproductive, household, and contextual confounders, women with an autonomy of decision made were revealed to have much lower odds of depression at the national level (AOR = 0.63; 95% CI: 0.39–0.98,  $p = 0.048$ ). Additionally, there was a significant rural-urban heterogeneity indicated by stratified analyses. Decision-making autonomy in rural areas was significantly associated to lower chances of having depressive symptoms (AOR = 0.44; 95% CI: 0.27–0.74,  $p = 0.002$ ) only [Table 2].

An AUC of 0.68 implies that, in approximately 68% of randomly selected case–non-case pairs, the model assigns a higher predicted probability to the woman with the outcome than to the woman without the outcome [Fig. 5].

Lower levels of autonomy were concentrated in the southern and south-western provinces (Quthing, Qacha's Nek, Mafeteng), which represent high-burden areas of limited decision-making power. In contrast, the central and northern provinces, including Thaba-Tseka and Maseru, exhibited relatively higher autonomy, although overall empowerment levels remained modest. These geographic patterns highlight meaningful regional variation in female autonomy and identify priority areas for context-specific interventions [Fig. 7].

The central-northern districts, especially the Maseru, Berea, Leribe, and Butha-Buthe, proved to have a higher prevalence of depression, which resulted in the identification of these districts as high-burden areas. Conversely, it was showed to be lower in the southern and south-

eastern districts, which comprise of Quthing, Qachas Nek, Mafeteng, and some parts of Hoek of Mohale which are relatively low-burden areas. These geographic inequalities reveal that there exist significant subnational differences in mental health burden and that mental health planning and resource allocation needs to be done at the district level with a focus on more burdensome central-northern districts [Fig. 6].

## 8. Discussion

Using nationally representative LDHS 2023–24 data, this study found that women in Lesotho experienced moderate-severe depressive symptoms, confirming a non-trivial mental health burden. Spatial analysis revealed substantial geographic inequalities: central-northern districts, including Maseru, Berea, Leribe, and Butha-Buthe, had the highest prevalence of depression, whereas southern and south-western districts (Quthing, Qacha's Nek, Mafeteng) exhibited the lowest autonomy and comparatively lower depression prevalence. These findings highlight the importance of women's decision-making autonomy for mental health, particularly in rural areas, and underscore spatial disparities that warrant targeted interventions. These findings are interpreted within the social determinants of mental health framework, highlighting those structural inequities, reproductive stressors, and decision-making autonomy shape women's vulnerability to depressive symptoms in Lesotho.

Previous studies have shown that depressive symptoms are a growing concern in Southern Africa, particularly among women facing social and economic vulnerabilities (Melkam et al., 2025; Mokwena and Makhozonke, 2024). Our study adds new evidence by linking decision-making autonomy directly to depressive symptoms at both national and spatial levels, highlighting rural-urban and geographic disparities (Noor et al., 2025). According to our findings, the sample proportion of rural residents was a little bit higher (55%), but the rates of depressive symptoms were slightly higher in women in the urban group. The observed rural-urban differences may reflect the distinct social and economic environments in Lesotho. Although rural women are not easily accessible to professional mental health care providers, their social and family networks provide them with more advantages in terms of making decisions and dealing with problems, increasing the protective value of autonomy (Acharya et al., 2010b; Ameyaw et al., 2016). On the other hand, even though urban women have more resources at their

**Table 1**Distribution of depressive symptoms (PHQ-9  $\geq 10$ ) by sociodemographic and reproductive characteristics (N = 3297).

| Characteristics                        | Depressive Symptoms ( $\geq 10$ ) %, (95% CI) |                   |
|----------------------------------------|-----------------------------------------------|-------------------|
|                                        | No                                            | Yes               |
| Overall                                | 92.6 (91.3, 93.7)                             | 7.4 (6.3, 8.7)    |
| <b>Women's age</b>                     |                                               |                   |
| 15–19                                  | 19.9 (18.2, 21.7)                             | 19.6 (13.4, 27.8) |
| 20–24                                  | 18.4 (16.7, 20.3)                             | 14.5 (9.3, 21.8)  |
| 25–29                                  | 13.5 (11.9, 15.3)                             | 16.2 (10.1, 24.9) |
| 30–34                                  | 13.6 (12.2, 15.1)                             | 13.2 (8.4, 20.1)  |
| 35–39                                  | 13.1 (11.6, 14.9)                             | 16.6 (11.1, 24.1) |
| 40–44                                  | 11.9 (10.4, 13.6)                             | 13.5 (8.9, 20.0)  |
| 45–49                                  | 9.6 (8.0, 11.5)                               | 6.5 (3.7, 11.1)   |
| <b>Age at first birth</b>              |                                               |                   |
| 10–17 (Early)                          | 11.4 (9.6, 13.4)                              | 17.1 (11.6, 24.5) |
| 18–24 (Normal)                         | 44.0 (41.8, 46.3)                             | 45.8 (38.1, 53.7) |
| $\geq 25$ (Late)                       | 44.6 (42.4, 46.8)                             | 37.0 (29.9, 44.8) |
| <b>Age at first sexual intercourse</b> |                                               |                   |
| < 18                                   | 54.4 (51.6, 57.2)                             | 64.0 (56.4, 70.9) |
| $\geq 18$                              | 45.6 (42.8, 48.4)                             | 36.0 (29.1, 43.6) |
| <b>Age at cohabitation</b>             |                                               |                   |
| < 18 (Young)                           | 14.2 (12.4, 16.1)                             | 19.4 (13.9, 26.3) |
| 18–20 (Mild)                           | 20.2 (18.5, 22.0)                             | 17.0 (11.7, 24.0) |
| $\geq 21$ (Older)                      | 65.6 (63.3, 67.9)                             | 63.7 (55.2, 71.3) |
| <b>Women's education</b>               |                                               |                   |
| Below Primary                          | 24.7 (22.1, 27.4)                             | 28.7 (21.0, 38.0) |
| Secondary incomplete                   | 59.2 (56.5, 61.9)                             | 54.1 (45.3, 62.6) |
| Secondary or higher                    | 16.1 (13.3, 19.5)                             | 17.2 (11.4, 25.2) |
| <b>Maternal Employment</b>             |                                               |                   |
| Not working                            | 61.0 (57.7, 64.2)                             | 61.6 (52.9, 69.6) |
| Currently working                      | 39.0 (35.8, 42.3)                             | 38.4 (30.4, 47.1) |
| <b>Adequate ANC Visits</b>             |                                               |                   |
| < 4 visits                             | 81.0 (79.1, 82.8)                             | 82.5 (75.5, 87.9) |
| $\geq 4$ visits                        | 19.0 (17.2, 20.9)                             | 17.5 (12.1, 24.5) |
| <b>Number of children</b>              |                                               |                   |
| None                                   | 33.6 (31.5, 35.8)                             | 26.7 (20.4, 34.2) |
| 1–2                                    | 48.2 (45.5, 51.0)                             | 48.5 (40.9, 56.1) |
| 3+                                     | 18.2 (16.4, 20.0)                             | 24.8 (18.8, 32.1) |
| <b>Birth Size</b>                      |                                               |                   |
| Small                                  | 81.5 (79.6, 83.2)                             | 78.7 (71.3, 84.7) |
| Average or Large                       | 18.5 (16.8, 20.4)                             | 21.3 (15.3, 28.7) |
| <b>Preterm birth</b>                   |                                               |                   |
| Term ( $\geq 9$ months)                | 91.8 (90.1, 93.2)                             | 90.9 (84.8, 94.7) |
| Preterm (< 9 months)                   | 8.2 (6.8, 9.9)                                | 9.1 (5.3, 15.2)   |
| <b>Currently Pregnant</b>              |                                               |                   |
| No                                     | 97.1 (96.2, 97.7)                             | 98.0 (94.7, 99.2) |
| Yes                                    | 2.9 (2.3, 3.8)                                | 2.0 (0.8, 5.3)    |
| <b>Pregnancy Status</b>                |                                               |                   |
| Single pregnancy                       | 67.7 (65.5, 69.8)                             | 75.3 (68.8, 81.0) |
| Multiple pregnancy                     | 32.3 (30.2, 34.5)                             | 24.7 (19.2, 31.2) |
| <b>Pressure to be pregnant</b>         |                                               |                   |
| No                                     | 93.3 (91.5, 94.7)                             | 86.3 (76.2, 92.6) |
| Yes                                    | 6.7 (5.3, 8.5)                                | 13.7 (7.4, 23.8)  |
| <b>History of pregnancy loss</b>       |                                               |                   |
| None                                   | 89.7 (88.0, 91.1)                             | 84.5 (78.1, 89.3) |
| One loss                               | 8.2 (6.8, 9.8)                                | 12.6 (8.2, 18.8)  |
| Two or more losses                     | 2.2 (1.6, 3.0)                                | 2.9 (1.3, 6.7)    |
| <b>Residing Status</b>                 |                                               |                   |
| Not residing with husband/partner      | 35.5 (32.0, 39.2)                             | 37.2 (26.6, 49.2) |
| Residing with husband/partner          | 64.5 (60.8, 68.0)                             | 62.8 (50.8, 73.4) |
| <b>Currently Abstaining</b>            |                                               |                   |
| No                                     | 93.6 (92.5, 94.5)                             | 94.6 (90.4, 97.1) |
| Yes                                    | 6.4 (5.5, 7.5)                                | 5.4 (2.9, 9.6)    |
| <b>Number of Unions</b>                |                                               |                   |
| Once                                   | 60.5 (58.0, 62.9)                             | 57.3 (48.0, 66.2) |
| More than once                         | 39.5 (37.1, 42.0)                             | 42.7 (33.8, 52.0) |
| <b>Menstruated last 6 weeks</b>        |                                               |                   |
| No                                     | 24.7 (22.8, 26.8)                             | 27.5 (20.6, 35.8) |
| Yes                                    | 75.3 (73.2, 77.2)                             | 72.5 (64.2, 79.4) |
| <b>Decision Making Autonomy</b>        |                                               |                   |
| No                                     | 52.1 (49.7, 54.5)                             | 56.6 (47.0, 65.7) |

**Table 1 (continued)**

| Characteristics                      | Depressive Symptoms ( $\geq 10$ ) %, (95% CI) |                   |
|--------------------------------------|-----------------------------------------------|-------------------|
|                                      | No                                            | Yes               |
| Yes                                  | 47.9 (45.5, 50.3)                             | 43.4 (34.3, 53.0) |
| <b>IPV Justification</b>             |                                               |                   |
| Does not justify IPV                 | 80.2 (77.8, 82.5)                             | 83.9 (77.7, 88.6) |
| Justifies IPV                        | 19.8 (17.5, 22.2)                             | 16.1 (11.4, 22.3) |
| <b>Household Sanitation facility</b> |                                               |                   |
| Unimproved                           | 35.4 (32.2, 38.8)                             | 30.6 (23.1, 39.3) |
| Improved                             | 64.6 (61.2, 67.8)                             | 69.4 (60.7, 76.9) |
| <b>Source of drinking water</b>      |                                               |                   |
| Unimproved source                    | 16.2 (13.7, 19.0)                             | 13.1 (8.8, 19.1)  |
| Improved source                      | 83.8 (81.0, 86.3)                             | 86.9 (80.9, 91.2) |
| <b>Place of residence</b>            |                                               |                   |
| Urban                                | 44.3 (40.7, 47.9)                             | 54.5 (46.1, 62.7) |
| Rural                                | 55.7 (52.1, 59.3)                             | 45.5 (37.3, 53.9) |
| <b>Administrative province</b>       |                                               |                   |
| Butha-Buthe                          | 6.2 (5.3, 7.2)                                | 6.7 (4.5, 9.8)    |
| Leribe                               | 17.8 (15.4, 20.5)                             | 19.5 (14.1, 26.4) |
| Berea                                | 15.0 (12.0, 18.6)                             | 18.4 (12.3, 26.6) |
| Maseru                               | 32.8 (28.8, 37.0)                             | 36.8 (28.3, 46.2) |
| Mafeteng                             | 6.5 (5.5, 7.5)                                | 5.3 (3.5, 8.0)    |
| Mohale's Hoek                        | 4.7 (4.0, 5.6)                                | 2.2 (1.1, 4.5)    |
| Quthing                              | 3.6 (3.1, 4.3)                                | 3.7 (2.3, 5.8)    |
| Qacha's Nek                          | 2.9 (2.4, 3.5)                                | 2.9 (1.6, 5.2)    |
| Mokhotlong                           | 4.4 (3.5, 5.4)                                | 1.4 (0.7, 2.7)    |
| Thaba-Tseka                          | 6.2 (5.4, 7.1)                                | 3.1 (1.7, 5.6)    |
| <b>Religion</b>                      |                                               |                   |
| Other                                | 65.3 (62.9, 67.6)                             | 63.3 (54.5, 71.4) |
| Roman Catholic                       | 34.7 (32.4, 37.1)                             | 36.7 (28.6, 45.5) |
| <b>Currently working</b>             |                                               |                   |
| No                                   | 61.0 (57.7, 64.2)                             | 61.6 (52.9, 69.6) |
| Yes                                  | 39.0 (35.8, 42.3)                             | 38.4 (30.4, 47.1) |
| <b>Wealth Status</b>                 |                                               |                   |
| Poor                                 | 31.3 (27.6, 35.2)                             | 24.9 (19.0, 31.9) |
| Middle class                         | 19.2 (16.9, 21.7)                             | 19.1 (13.2, 26.9) |
| Rich                                 | 49.4 (45.2, 53.6)                             | 55.9 (47.1, 64.3) |
| <b>Household Size</b>                |                                               |                   |
| 1–3 members                          | 28.3 (25.6, 31.1)                             | 21.7 (15.9, 28.9) |
| 4 or more members                    | 71.6 (68.8, 74.3)                             | 78.2 (71.0, 84.0) |
| <b>Household head sex</b>            |                                               |                   |
| Male                                 | 57.6 (54.9, 60.2)                             | 55.4 (45.6, 64.8) |
| Female                               | 42.3 (39.7, 45.0)                             | 44.5 (35.1, 54.3) |
| <b>Household assets</b>              |                                               |                   |
| No                                   | 51.3 (47.3, 55.3)                             | 49.2 (41.6, 56.8) |
| Yes                                  | 48.6 (44.6, 52.6)                             | 50.7 (43.1, 58.3) |
| <b>Media Access</b>                  |                                               |                   |
| No                                   | 5.7 (4.77, 6.90)                              | 3.2 (1.79, 5.83)  |
| Yes                                  | 94.2 (93.0, 95.2)                             | 96.7 (94.1, 98.2) |
| <b>Internet use status</b>           |                                               |                   |
| No internet use                      | 19.6 (17.7, 21.6)                             | 22.0 (14.7, 31.4) |
| Internet use last 12 months          | 80.3 (78.3, 82.2)                             | 77.9 (68.5, 85.2) |
| <b>Household materials</b>           |                                               |                   |
| Unimproved                           | 17.0 (14.9, 19.2)                             | 11.5 (7.8, 16.7)  |
| Improved                             | 82.9 (80.7, 85.0)                             | 88.4 (83.2, 92.1) |

disposal, they are more prone to economic hardship, social isolation, and urban stress, all of which reduce the protective nature of autonomy on depression (Antabe et al., 2025). This is contrary to other LMIC studies where urban women can usually report improved mental health outcomes because of increased socioeconomic resources (Asmare et al., 2026). However, in Lesotho, urban life can be accompanied by a greater exposure to stressors, including economic, social isolation or urban poverty, which may at least partly be responsible for the observed urban vulnerability (Mokwena and Makhozonke, 2024). It was determined that events and pressures in early life including the earlier age of first birth, earlier sexual initiation, earlier cohabitation, and pregnancy loss were connected with increased depressive symptomatology (Moeti et al., 2022; Russell et al., 2021). These results are in line with the previous studies that associate reproductive stressors and early life transitions to mental health vulnerability in the long term as a woman (Bushu

**Table 2**  
Associations Between Decision Making Autonomy and Depression Symptoms (N = 3297, N1 Rural = 2086, N2 Urban = 1211).

| Characteristics          | Depression (PHQ-9 ≥ 10) |                        |                       |                        |
|--------------------------|-------------------------|------------------------|-----------------------|------------------------|
|                          | COR (95% CI), p         | AOR (95% CI),p         |                       |                        |
| Decision Making Autonomy | -                       | National Level         | Rural                 | Urban                  |
| No (Ref)                 | 1                       | 1                      | 1                     | 1                      |
| Yes                      | 0.83 0.56–1.23),0.368   | 0.63(0.39–0.98), 0.048 | 0.44(0.27–0.74),0.002 | 0.88 (0.39–1.97),0.758 |

Note: OR = odds ratio; CI = confidence interval; PHQ-9 ≥ 10 indicates depression; GAD-7 ≥ 10 indicates anxiety.  
Confounders: women age, women education, maternal employment, religion, administrative division, age at first birth, age at cohabitation, number of children, pregnancy status, pregnancy loss, place of residence, toilet facilities, water access, household head sex, household size category, household wealth status, media access, internet use, household assets, and household materials.

et al., 2025). It is important to note that the association between the depressive symptoms and pregnancy loss highlights the urgent necessity of providing women with reproductive difficulties with a specific kind of psychosocial assistance (Boakye et al., 2025).  
Taken together, these findings suggest that empowering women in decision-making and addressing early-life reproductive stressors are critical for reducing depressive symptoms. The implications for policy include designing context-specific, equity-oriented interventions that combine reproductive and mental health support, with particular attention to rural areas and high-burden districts.  
Less education and unemployment and poor household wealth were also associated with it but with slightly high prevalence of depression, though it was within the same confidence interval (Quenby et al., 2021). This is in line with the current social determinants of mental health framework that identifies structural inequities as a way of aggravating the risk factors on the individual level (Alkema et al., 2016). In the national level, autonomy in making decisions showed lower odds

with depression, and this association was even stronger in rural settings. The results are consistent with other studies on Southern Africa that showed women empowerment and autonomy to be protective against depressive symptoms (Miller et al., 2016). The predictive performance of the multivariable model was modest around (AUC = 0.68), indicating fair discrimination. While the model performs better than chance, it does not achieve a level of accuracy sufficient for individual-level prediction. This is consistent with prior population-based mental health studies, where complex psychosocial outcomes are influenced by unmeasured factors such as interpersonal relationships, trauma, and social norms (Ostovar et al., 2025). As another finding of our spatial analyses, central-northern districts showed more depression prevalence whereas southern and south-western provinces (Quthing, Qacha Nek, Mafeteng, Mophale Hoek) showed more autonomy and less depression (DHS Program, 2023; Mokwena and Makhozonke, 2024). These trends can be aligned with the hypothesis that geographic disparities in enabling and psychosocial stress are associated with unequal mental health workloads in districts (Thibaut and ELNahas, 2023). Interestingly, health-related variables like antenatal care, parity and multiple pregnancies were found to have minimum dependence on depression whereas media and internet exposure did not seem to have any effects on the symptom patterns (Flor et al., 2026). This is contrary to other LMICs where utilization of health services tends to mediate the mental health outcomes implying that the context of social and other determinants of mental health may play a higher role in the Lesotho context (Miah and Ullah, 2025; Noor, Siddique, Das et al., 2025; Pandey et al., 2024). Altogether, these results demonstrate the need of context-specific, equity-oriented interventions, which combine both reproductive and mental health support (Zhang et al., 2019). The observed spatial variations, with higher depression prevalence in central–northern districts and lower prevalence in southern and south-western provinces, are likely due to geographic differences in women’s autonomy, socio-economic resources, psychosocial stressors, and urban-rural contextual factors. These findings revealed the need for geographically targeted mental health interventions (Abedin and Arunachalam, 2020). The identified geographic and sociodemographic differences also suggest that national mental health policy must focus on central-northern dis-

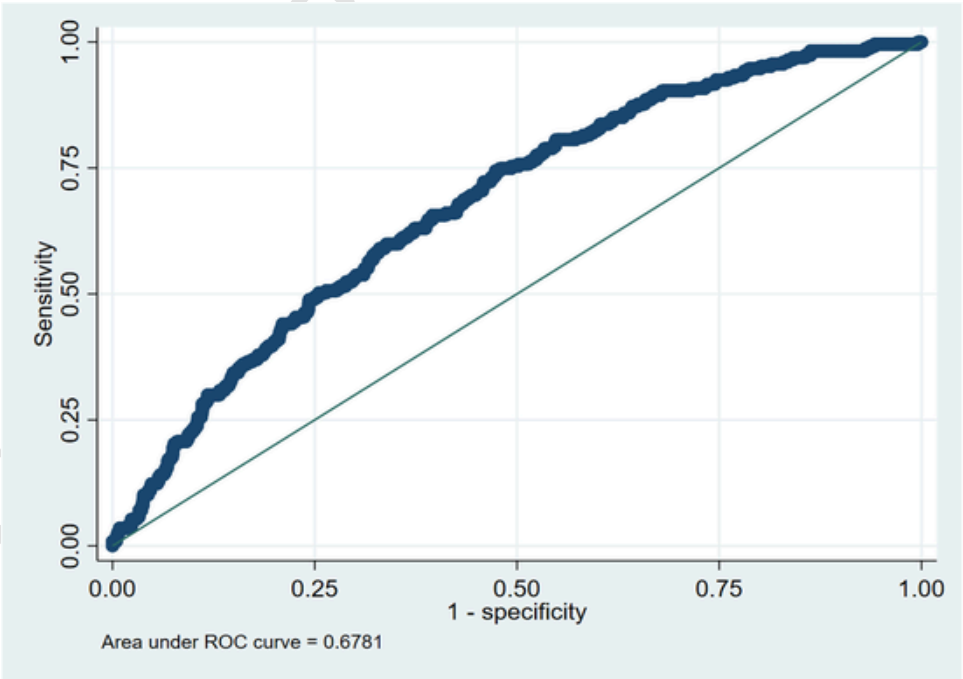


Fig. 5. ROC curve showing model performance.

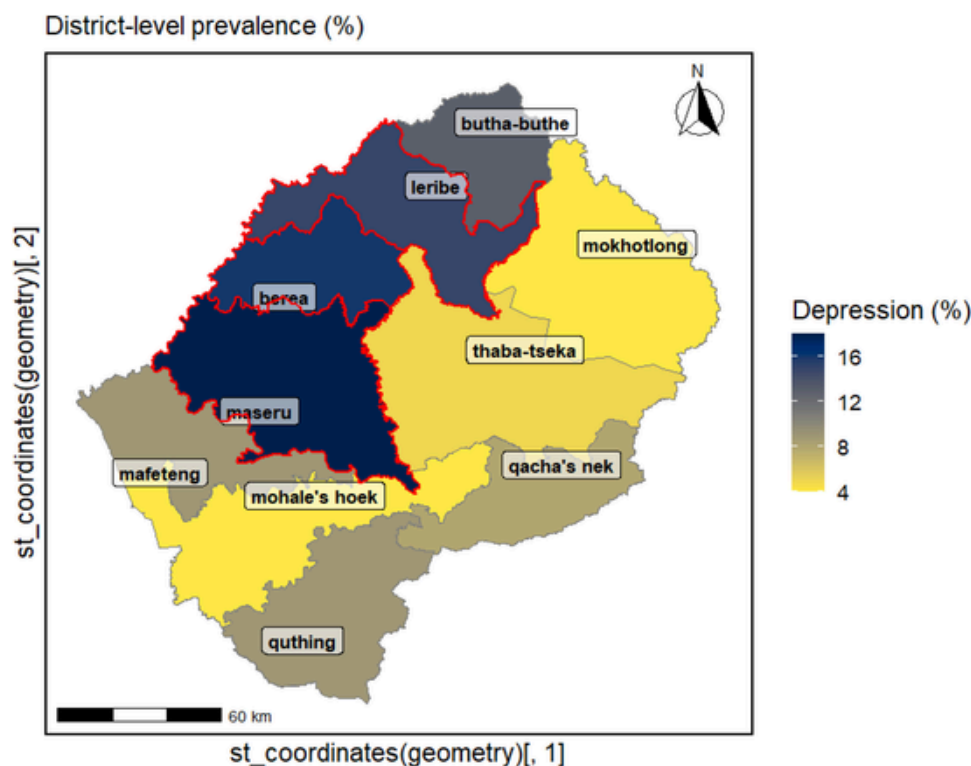


Fig. 6. Spatial Inequalities of Depression Prevalences.

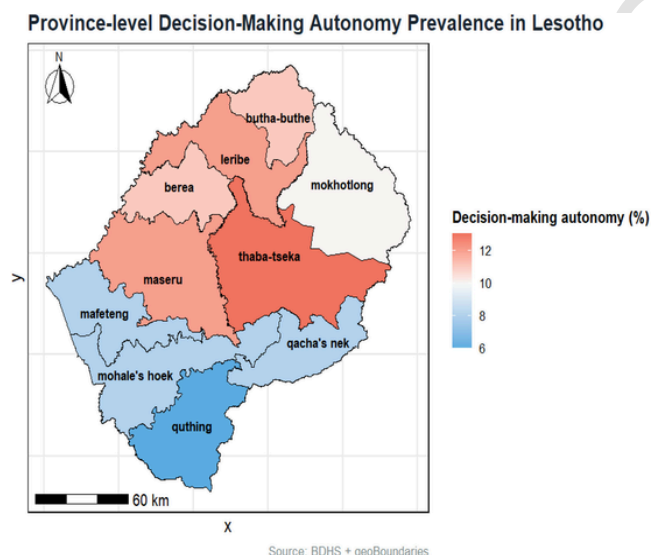


Fig. 7. Spatial Inequalities of Decision-making Autonomy.

districts and women with early-life reproductive stressors, as well as encourage autonomy in decision-making as a protective factor (Llop-Gironés et al., 2019). The combination of spatial and individual-level analysis in this research offers practical piece of evidence in order to guide the allocation of resources and specific mental health programming in Lesotho (Dangare et al., 2026).

Overall, this study demonstrates what was previously known, adds novel insights into the protective role of autonomy and geographic disparities, and underscores the need for tailored, empowerment-focused mental health interventions in Lesotho.

## 9. Policy implications and recommendations

The findings indicate that mental health services in Lesotho need to be provided in context-specific manner. Policies should: Empower rural women who have a low autonomy, improve empowerment initiatives, reproductive decision-making support, and interventions aimed at psychosocial. Increase the central-northern districts with high burden of mental health in need of expanded services, community awareness, and primary care integration. Integrate the empowerment strategies into the wider maternal and reproductive health programs, recognizing the differing impacts between the rural and the urban areas.

## 10. Strengths and Limitations

The strengths are the utilization of national representative data, the measurement of depression based on PHQ-9, and the incorporation of the spatial analysis that can be used to demonstrate geographic inequities. This study has several limitations, such as its cross-sectional design, which excludes the possibility of causal inference, and use of self-reported measures of autonomy and depression, which is prone to reporting bias. In spite of these shortcomings, the article presents strong evidence on how the issues of empowerment, reproductive factors and mental health intersect in Lesotho. Additionally, the PHQ-9 relies on self-reported symptoms and may not fully capture clinical depression, and its sensitivity and specificity can vary across cultural and demographic contexts. Future studies may consider alternative PHQ-9 cut-offs or item-level sensitivity checks to further assess robustness.

## 11. Conclusions

Despite these limitations, the study revealed strong evidence on the association between decision making autonomy and depressive symptoms with rural-urban disparities in Lesotho. These results indicate that the autonomy of women can be a protective factor against depression in

rural areas, where the structural and social limitations to the agency of women may be higher. On the whole, the findings reveal context-specific mental health gains of women empowerment and the need to incorporate in rural maternal and mental health interventions autonomy-enhancing strategies. The results indicate that there exist high rural-urban inequalities, geographic inequalities, central-northern districts had prevalence of depression and lowest autonomy was observed in southern-southwestern districts. These findings highlight the importance of context-sensitive, empowerment-based mental health services, especially the rural women and high-burden districts. The combination of women empowerment strategies with reproductive health and mental health initiatives might help to decrease the symptoms of depression and help to ensure equality in mental health outcomes nationwide.

## Data availability

The data used in this study are publicly available from the Demographic and Health Surveys (DHS) Program and can be requested at <https://www.dhsprogram.com/data>.

## Assistance of generative AI tools

AI-assisted tools including Grammarly, QuillBot, and ChatGPT were used solely for language editing and grammar improvements. AI did not contribute to data analysis, interpretation, or intellectual content. The authors verified the manuscript for originality and ethical compliance.

## Consent to participant

Not applicable.

## Consent for publication

Not applicable. This study used de-identified, publicly available data and does not include any identifiable information.

## Funding

This study did not receive any funding from public, private, or non-profit organizations.

## CRediT authorship contribution statement

**Md Salek Miah:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria Bintey Kabir:** Supervision, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare no competing interests.

## Acknowledgements

We thank the DHS Program for providing free access to the 2022 survey data. We also acknowledge Dr. Jibanul Haque (PhD, University of Central Florida; AI Scientist, The Home Depot) for guidance and support in designing, and implementing, which improved the rigor and clarity of this study.

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